

Electrocardiogram data compression using DCT based discrete orthogonal Stockwell transform

Chandan Kumar Jha*, Maheshkumar H. Kolekar

Department of Electrical Engineering, Indian Institute of Technology Patna, 801103, India

ARTICLE INFO

Article history:

Received 27 September 2017

Received in revised form 14 June 2018

Accepted 27 June 2018

Available online 29 July 2018

Keywords:

Discrete orthogonal Stockwell transform

ECG

Dead zone quantization

Compression ratio

Percent root mean square difference

Quality score

ABSTRACT

This paper reports a novel electrocardiogram (ECG) data compression algorithm which employs DCT based discrete orthogonal Stockwell transform. Dead-zone quantization is utilized to apply quantization as well as a threshold condition to transform coefficients. Further, integer conversion of coefficients is performed. It improves compression at the cost of very less reconstruction error. All integer coefficients are encoded using run-length coding. It exploits the repetition of data instances. Run-length coding helps to achieve higher compression without any relevant information loss. Performance of the proposed compression algorithm is evaluated using 48 single channel ECG records which are taken from the MIT-BIH arrhythmia database. A competitive compression performance is observed in comparison with other ECG compression methods.

© 2018 Elsevier Ltd. All rights reserved.

1. Introduction

Electrocardiogram (ECG) is a graphical recording of tiny electrical impulses generated by heart muscles. It is used extensively in cardiology. Cardiologists use the ECG signal as a non-invasive medical tool to detect abnormalities in the human heart. In the long-term monitoring of heart-patients, recorded ECG data acquire a large volume of memory space for storage. It also consumes large bandwidth of communication channels in case of transmission of data for remote monitoring of patients or telemedicine. In this regard, application of ECG compression techniques resolves the problems of a healthcare system used for remote monitoring, Holter monitoring, patient-history review or telemedicine.

In past, many ECG compression techniques have been proposed which are broadly classified into three types [1]: direct, parameter extraction and transform domain methods. Direct methods remove redundancy of ECG signal directly in time-domain. This class includes turning point (TP), amplitude zone time epoch coding (AZTECH) [2], improved AZTEC coding [3], entropy coding [4] and ASCII character encoding [5]. The parameter extraction methods are based on the extraction of important features of ECG signal such as P-wave section, QRS-complex, and T-wave section. Extracted features utilize beat codebook matching and long-term prediction [6] to achieve compression. Other typical examples

of this method include artificial neural network [7], peak picking and vector quantization [8]. In transform domain methods, first ECG signal is decomposed by means of a linear orthogonal transformation, after which the transform coefficients are appropriately encoded. Discrete Fourier transform (DFT) [9], discrete cosine transform (DCT) [10], Walsh transform [11], discrete wavelet transform (DWT) [12–16] are commonly used transform methods for ECG data compression. Numerous researchers have proposed ECG compression techniques based on two-dimensional transform methods such as 2D-DCT [17], 2D-DWT [18,19] and singular value decomposition (SVD) [20]. Empirical mode decomposition [21,22] and accuracy driven sparse model [23] were recently introduced for ECG data compression.

In the field of ECG data compression, transform-based techniques have gained a significant growth in the past few decades. Among transform based ECG compression approaches, Fourier transforms (FTs) and wavelet transform are used widely. Since FTs are not able to provide information regarding the time occurrence of frequency components of the signal. Therefore, it is not suitable for the analysis of non-stationary signals. Limitations of FTs can be overcome by introducing a constant window in short-time Fourier transform (STFT) but selection of window size is a problem. Wavelet transform fixed this problem by selecting a variable window size. Larger window is used at low frequency and shorter window at high frequency of the signal. In wavelet transform, proper selection of sampling frequency and mother wavelet plays a major role in frequency component extraction, failing which may produce misleading information. The drawback of wavelet trans-

* Corresponding author.

E-mail addresses: ckjha.pee15@iitp.ac.in, ckjha.bit@gmail.com (C.K. Jha).

form is overcome by Stockwell transform (S-transform) [24] which combines the frequency-dependent resolution of time-frequency space and local phase information uniquely. The phase information is absolutely referenced. Till now, S-transform has been used in many research areas such as atmospheric studies, cardiovascular studies, magnetic resonance image analysis, characterization of seismic signals etc. In this paper, S-transform is employed as a compression tool for ECG signal which is a novel aspect in the area of ECG signal compression.

Organization of this paper is as follows: Section 2 introduces the discrete orthogonal Stockwell transform. Methodology of the proposed technique is elaborated in Section 3. Section 4 describes the performance metrics. Experimental results and discussion are detailed in Section 5. Section 6 presents the concluding remarks of this paper.

2. Discrete orthogonal Stockwell transform (DOST)

S-transform bridges the gap between FT and wavelet transform. It uniquely combines the time-frequency resolution with absolutely referenced phase (the phase referenced to time $t=0$) information. As described in [24], continuous S-transform of an input function $h(t)$ can be defined as the product of FT of $h(t)$ and a Gaussian window.

$$S(\tau, f) = \frac{|f|}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} h(t) e^{-(\tau-t)^2 f^2 / 2} e^{-i2\pi f t} dt \quad (1)$$

$S(\tau, f)$ is a one-dimensional function of time for a constant frequency f . It shows how the amplitude and phase change over time for the frequency f . If $h[kT]$, $k=0, 1, 2, \dots, N-1$ is the discrete form of $h(t)$ and $H[n/NT]$ is the discrete Fourier transform (DFT) of $h[kT]$ then discrete S-transform of $h[kT]$ can be represented as

$$S \left[jT, \frac{n}{NT} \right] = \sum_{m=0}^{N-1} H \left[\frac{m+n}{NT} \right] e^{-2\pi^2 m^2 / n^2} e^{i2\pi m j / N} \quad (2)$$

where T is the sampling time interval, $n=1, 2, 3, \dots, N-1$ and $j=m=0, 1, 2, \dots, N-1$. In the discrete case, S-transform is suffered from the high redundancy of data instances. It represents a signal of length N by N^2 data instances. DOST is developed to remove the redundancies of discrete S-transform which has an orthonormal basis and multiple scales. An orthonormal transformation shows an N -point time-frequency representation of an N -point time series, thus maximum representation efficiency can be achieved by using it.

The efficient representation of DOST can be defined as the inner product of time series $h[kT]$ and the basis functions defined in terms of the function of $[kT]$. Parameters of the basis functions are ν , β , and τ . ν is a frequency variable which indicates the centre of the frequency band. β indicates the width of the frequency band and τ is a time-variable which denotes the time localization.

$$S(h[kT]) = S \left(\tau T, \frac{\tau}{NT} \right) = \sum_{k=0}^{N-1} h[kT] S_{[\nu, \beta, \tau]}[kT] \quad (3)$$

In general case, basis functions $S_{[\nu, \beta, \tau]}[kT]$ are defined as

$$S_{[\nu, \beta, \tau]}[kT] = \frac{ie^{-i\pi\tau}}{\sqrt{\beta}} \frac{e^{-i2\pi(k/N - \tau/\beta)(\nu - \beta/2 - 1/2)} - e^{-i2\pi(k/N - \tau/\beta)(\nu + \beta/2 - 1/2)}}{2 \sin[k/N - \tau/\beta]} \quad (4)$$

At this point, orthogonality is ensured by applying some rules to the sampling of time-frequency space. These rules are as follows:

- 1.) Rule 1: $\tau = 0, 1, 2, \dots, \beta - 1$
- 2.) Rule 2: ν and β have to be chosen suitably such that each Fourier frequency sample is used once.

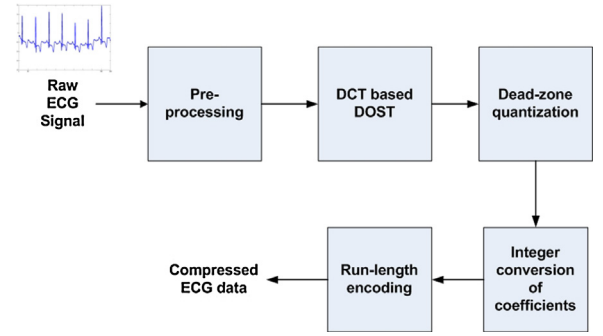


Fig. 1. Block diagram of compression procedures.

For each frequency band (β), there are one or more local time samples (τ). It should be equal to β (by Rule 1). Thus, wider frequency resolution (large β) results in more samples in time (large τ). Values of ν and β are determined by imposing some specific rules on the basis function. Octave sampling [24] is used for the strict definition of ν and β . Operation of DOST can be summarized as:

Algorithm 1. Operation of DOST

INPUT: discrete time series $h[kT]$

OUTPUT: DOST of $h[kT]$

Step 1: Perform DOST by the inner product of $h[kT]$ and $S_{[\nu, \beta, \tau]}[kT]$

Step 2: To ensure orthogonality, apply the two aforementioned rules to τ , ν and β . For this employ octave sampling to time-frequency space.

3. Proposed methodology

Methodology of the proposed compression algorithm is shown in Fig. 1 using a block diagram. It involves following steps i.e., pre-processing of raw ECG signal, DOST implementation, dead-zone quantization of DOST coefficients, integer conversion of coefficients and run-length encoding.

3.1. Pre-processing

At first, raw ECG signals are acquired from the MIT-BIH arrhythmia database [25]. Base and gain of raw ECG signals are 1024 and 200 respectively. Original ECG signal is achieved from the raw ECG signal by subtracting base and dividing the resulting data by gain. Much high frequency (HF) noises are present in the original ECG signal due to artifacts such as muscle contraction, power-line interference, and electrode movement. These HF noises are removed by employing Savitzky–Golay filter (SGF). SGF is also known as polynomial smoothing filter which rejects HF noises efficiently without losing key information of the signal. Polynomial order 3 and window dimension 17 are chosen for SGF.

3.2. DCT based DOST

After filtering, DOST is applied to the ECG data. Instead of DFT, discrete cosine transform (DCT) is utilized in the kernel of the DOST. DCT is a real-valued transform which shows energy compaction property. It is widely used in the field of data compression. The DCT based DOST produces only positive (increasing) frequencies [26] and carries no symmetry in the coefficients. Consequently, higher frequencies are required in frequency space. There is also a need of adjustment in the partitioning of frequency space. However, it is achieved by continuing dyadic partition as is used in the case of DOST.

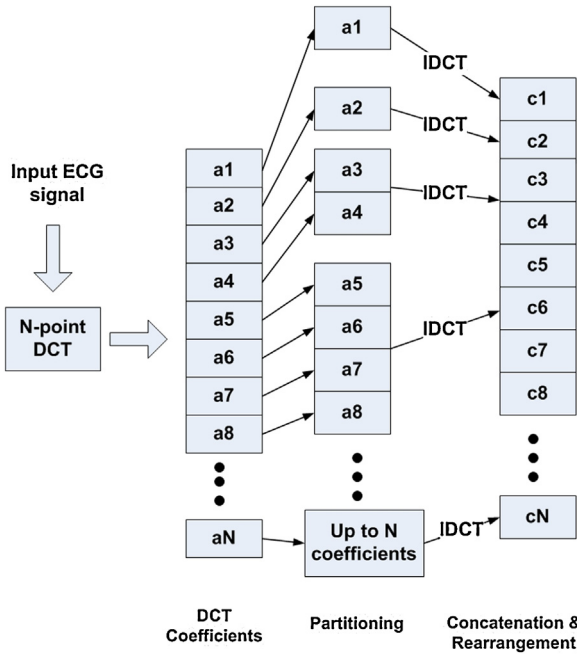


Fig. 2. Block diagram of DCT based DOST implementation.

For a given signal of length $N=2^M$, a variable p is defined such that

$$p = 1, 2, 3, \dots, \log_2 N + 1 = 1, 2, 3, \dots, M + 1 \quad (5)$$

where M is an integer. The parameters ν , β and τ , are determined based on the values of p . For $1 < p \leq M + 1$,

$$\nu = 2^{p-2} + 2^{p-3} \quad (6)$$

$$\beta = 2^{p-2} \quad (7)$$

$$\tau = 0, 1, 2, \dots, 2^{p-2} - 1 \quad (8)$$

For the case, $p=1$ then $\nu=1$, $\beta=1$ and $\tau=0$.

The complete procedure of DCT based DOST implementation is shown in Fig. 2 using a block diagram. Initially, N -point DCT is applied to the input ECG signal of length $N=2^M$. It generates DCT coefficients which are shown in Fig. 2 as $a_1, a_2, a_3, a_4, \dots, a_N$. The obtained coefficients are partitioned into frequency sub-bands. According to length, the number of coefficients in frequency sub-bands is determined as $[1, 2^0, 2^1, 2^2, \dots, 2^{M-1}]$. Further, β -point inverse DCT is employed to each frequency sub-band to compute the coefficients which are localized in space. After computation, resulting coefficients ensure the decomposition is orthogonal. In Fig. 2, β -point inverse DCT coefficients are represented as $c_1, c_2, c_3, c_4, \dots, c_N$. These coefficients are treated as DCT based DOST coefficients. For time-frequency analysis, these coefficients are concatenated and rearranged. Partitioning of time-frequency space is performed based on τ and β values while ν is used to denote the centre of the frequency band.

The DCT based DOST works well on ECG signal whose length is in power of two. However, this should not be a strict requirement. By truncating or zero padding, the DCT based DOST can be applied to any arbitrary length of ECG signal.

3.3. Dead-zone quantization

Dead-zone quantization (DZQ) is employed to all obtained transform coefficients. The DZQ scheme [27] can be described as follows:

$$D_k = \begin{cases} (-3\delta, -T_h], & \text{if } k = -1 \\ (-T_h, T_h), & \text{if } k = 0 \\ [T_h, 3\delta), & \text{if } k = 1 \\ [(2k-1)\delta, (2k+1)\delta), & \text{otherwise} \end{cases} \quad (9)$$

$$R_k = \begin{cases} 0, & \text{if } k = 0 \\ k\Delta, & \text{for } k = \pm 1, \pm 2, \pm 3, \dots \end{cases} \quad (10)$$

where D_k is the k th decision interval for input DOST coefficients and R_k is the corresponding quantized output value. Δ is the quantization step size while δ represents half of the quantization step size. In DZQ scheme, a small interval $(-T_h, T_h)$ is chosen around zero and all small DOST coefficients falling in that interval are converted to zero. T_h is called the threshold value of the DZQ and the interval $(-T_h, T_h)$ is known as dead-zone whose width is $2T_h$ ($\Delta < 2T_h$). Energy packing efficiency (EPE) [28] is employed to determine the threshold value T_h . EPE can be defined as:

$$EPE = \frac{E_{Th}}{E} \times 100\% \quad (11)$$

where E and E_{Th} denote energy of the transform coefficients before and after thresholding. In DZQ, except dead-zone and adjacent intervals for $k=1$ and $k=-1$, all other decision intervals are of equal width 2δ . Step size Δ is determined by the relation $\Delta = \alpha T_h$. From [29], the prescribed range of α is 1.20 to 1.80. For all ECG records used for experimentation, $\alpha = 1.60$ is the satisfactory choice which offers good compression performance.

3.4. Integer conversion of coefficients

After performing DZQ of DOST coefficients, most generated coefficients values (except coefficients lying in dead-zone) are in fractions of the order of 10^{-5} . The coefficients require double data format (64 bits) to store it in computer memory. To reduce the number of bits requirement, all coefficients are multiplied by 10^4 and rounded-off to convert them in integer values. After integer conversion, all coefficients are saved in 16-bit signed integer format. This step improves compression at the cost of very less reconstruction error.

3.5. Run-length encoding

Further, run-length encoding (RLE) [30] is utilized to exploit the repetition of data instances. It represents the data as run and length. RLE enhances compression without any loss of information. The encoded data is the compressed form of original ECG signal.

Original ECG signal is reconstructed from the compressed data by following the inverse procedure of compression. First, the compressed ECG data is decoded using inverse RLE. The decoded data is converted in double data format and divided by 10^4 (because the data is multiplied by 10^4 during integer conversion of coefficients in compression procedure). Finally, the inverse DCT based DOST is employed to obtain the reconstructed ECG signal.

4. Performance metrics

Performance of the proposed compression algorithm is evaluated based on the following metrics which are widely used in the literature [10].

4.1. Compression ratio (CR)

CR indicates the ability of the compressor to eliminate the redundant data. If the CR is higher, number of bits required to store or transmit the data is fewer. CR is defined as follows:

$$CR = \frac{\text{Size of the original ECG signal}}{\text{Size of the compressed ECG signal}} \quad (12)$$

4.2. Percent root mean square difference (PRD)

PRD is a measure of the degree of distortion between the original and reconstructed ECG waveforms. It is defined as follows:

$$PRD = \sqrt{\frac{\sum_{i=1}^N (x_i - x_i^r)^2}{\sum_{i=1}^N x_i^2}} \times 100 \quad (13)$$

where x_i and x_i^r are the original and reconstructed ECG data respectively.

4.3. Normalized percent root mean square difference (PRDN)

It is the normalized version of PRD. PRDN does not depend on the mean value $\bar{X}$ of the signal.

$$PRDN = \sqrt{\frac{\sum_{i=1}^N (x_i - x_i^r)^2}{\sum_{i=1}^N (x_i^2 - \bar{X}^2)}} \times 100 \quad (14)$$

4.4. Quality score (QS)

It is the ratio of CR and PRD. QS is an important performance measure which is used to determine the best compression method while taking into account of compromised reconstruction errors [10]. In a lossy compression method, a high QS value represents a high quality of compression.

$$QS = \frac{CR}{PRD} \quad (15)$$

4.5. Root mean square error (RMS)

RMS error is a distortion measure with respect to sample size. It is defined as:

$$RMS = \sqrt{\frac{\sum_{i=1}^N (x_i - x_i^r)^2}{N - 1}} \quad (16)$$

4.6. Signal to noise ratio (SNR)

Due to compression and decompression procedure, some noise energy is introduced in the signal. SNR is the measure of the degree of noise energy in decibel (dB) scale.

$$SNR = 10 \times \log\left(\frac{\sum_{i=1}^N (x_i - \bar{X})^2}{\sum_{i=1}^N (x_i - x_i^r)^2}\right) \quad (17)$$

5. Results and discussion

For evaluation of the compression performance of the proposed algorithm, all 48 ECG records of the MIT-BIH arrhythmia database [25] are used for experimentation. These records are sampled at 360 Hz with the resolution of 11-bits/sample. In terms of number of samples, length of each ECG record taken is 8192. In DZQ, threshold values for different ECG records are determined with $EPE = 99.999\%$. Table 1 provides the compression results for all 48 ECG records. Average CR, PRD, PRDN, QS, RMS, and SNR for all records are 6.27,

Table 1

Compression performance for ECG records of the MIT-BIH arrhythmia database.

ECG record no.	Performance metrics					
	CR	PRD	PRDN	QS	RMS	SNR
100	5.74	3.72	8.43	1.54	0.01	21.49
101	5.80	3.99	7.45	1.45	0.01	22.55
102	4.92	3.33	5.54	1.48	0.01	25.12
103	7.16	6.33	8.42	1.13	0.02	21.49
104	6.33	5.44	7.28	1.16	0.02	23.76
105	6.38	5.41	6.95	1.18	0.02	23.16
106	6.99	8.74	9.76	0.80	0.04	20.21
107	6.61	10.69	11.23	0.62	0.10	18.99
108	5.10	2.61	5.15	1.96	0.01	25.76
109	5.68	3.32	4.51	1.71	0.02	26.92
111	5.61	4.22	5.96	1.33	0.01	24.50
112	4.84	1.02	4.76	4.76	0.01	26.45
113	6.76	5.45	6.18	1.24	0.02	24.17
114	4.82	2.48	5.21	1.94	0.01	25.66
115	6.98	4.73	9.91	1.48	0.03	20.74
116	7.91	12.82	24.74	0.62	0.14	12.13
117	5.45	1.30	4.83	4.20	0.01	26.33
118	6.19	1.83	5.56	3.38	0.02	25.10
119	8.15	10.83	21.93	0.75	0.10	13.18
121	6.87	2.06	4.97	3.33	0.02	26.07
122	6.27	2.24	5.99	2.80	0.02	24.45
123	6.53	2.60	7.46	2.51	0.01	22.55
124	7.80	4.88	12.05	1.60	0.04	18.38
200	6.78	7.15	7.54	0.95	0.03	22.45
201	5.56	4.98	6.96	1.12	0.01	23.14
202	6.87	6.86	7.39	1.00	0.02	22.63
203	6.15	5.57	5.93	1.10	0.03	24.54
205	5.23	3.15	7.06	1.66	0.01	23.03
207	6.17	4.00	4.55	1.54	0.01	26.85
208	6.22	6.49	6.65	0.96	0.03	23.55
209	5.81	5.57	6.79	1.04	0.01	23.37
210	6.06	5.16	6.24	1.17	0.01	24.10
212	5.40	4.51	5.04	1.20	0.02	25.95
213	6.71	8.57	11.01	0.78	0.06	19.16
214	7.16	7.29	7.86	0.98	0.04	22.09
215	5.86	5.62	6.24	1.04	0.01	24.09
217	6.27	4.85	4.88	1.29	0.03	26.23
219	7.38	8.36	14.50	0.88	0.06	16.77
220	6.76	3.57	7.88	1.89	0.02	22.07
221	6.34	5.26	6.22	1.20	0.02	24.12
222	5.55	5.02	6.59	1.10	0.01	23.63
223	6.79	4.92	8.84	1.38	0.03	21.08
228	7.33	10.44	12.67	0.70	0.04	17.95
230	6.13	6.37	7.76	0.96	0.03	22.20
231	5.74	4.76	5.27	1.21	0.01	25.56
232	4.88	3.53	7.25	1.38	0.01	22.79
233	6.60	9.86	10.45	0.67	0.06	19.62
234	6.42	5.91	6.66	1.08	0.02	23.53
Avg.	6.27	5.37	7.95	1.49	0.03	22.68

5.37, 7.95, 1.49, 0.03 and 22.68 respectively. For visual assessment of the proposed compression method ECG record no. 100 and 117 are selected. These records are widely used in the literature [31,12] for validation purpose. Therefore, these records are chosen. Original and reconstructed signals for record no. 100 and 117 are depicted in Figs. 3 and 4 respectively. By visual inspection, it is noticed that reconstructed signals are quite similar to original signals. R-peaks are an important feature of ECG signal. For quality assessment, R-peaks are detected in original and reconstructed signals of ECG record 100 and 117. The benchmark Pan and Tompkins algorithm [32] is employed to detect R-peaks. From Figs. 5 and 6, it is found that the detected R-peaks in original and reconstructed signals are at the same positions. Normalized frequency vs time plots are also shown in Figs. 8 and 9. In these figures, positive frequencies are shown from bottom to top and colours represent values of DOST coefficients. Analysis of these figures illustrates that time-frequency characteristics of reconstructed signals are very much similar to original ECG signals.

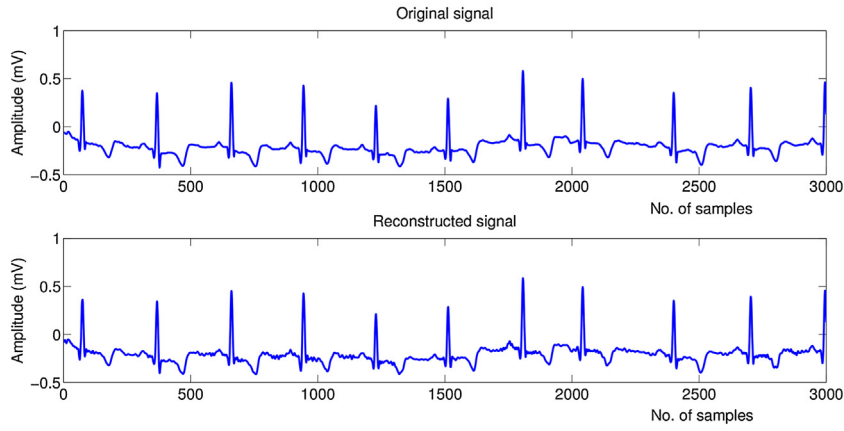


Fig. 3. Original and reconstructed ECG signals for record no. 100 of the MIT-BIH arrhythmia database (CR 5.74, PRD 3.72).

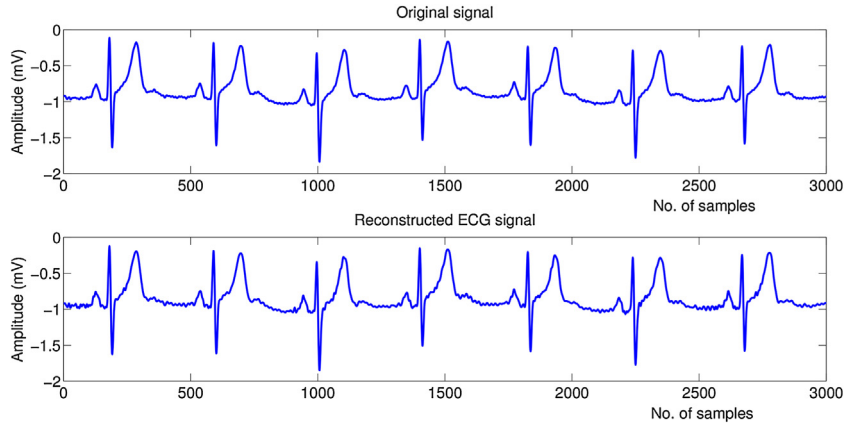


Fig. 4. Original and reconstructed ECG signals for record no. 117 of the MIT-BIH arrhythmia database (CR 5.45, PRD 1.30).

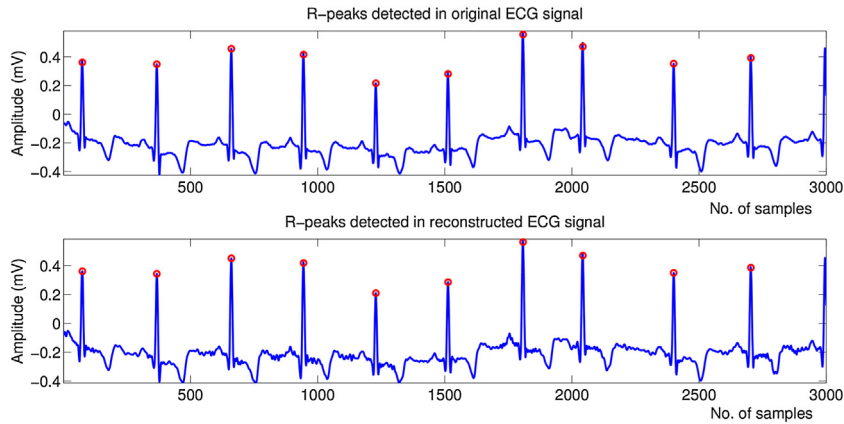


Fig. 5. R-peaks detected in original and reconstructed signals for ECG record no. 100 of the MIT-BIH arrhythmia database.

However, to assure the good diagnostic features of the reconstructed ECG signal, a subjective measure is necessary. It provides the true quality of the reconstructed signal. For this, a semi-blind mean opinion score (MOS) [33,34] test is carried out by four evaluators. Two medical doctors and two researchers, who are working in the field of biomedical signal processing, evaluated the reconstructed signal by comparing it to the original signal. In semi-blind MOS test, evaluators were not acquainted with the technique used for compression while they were aware of original and reconstructed signals. For the evaluation of the true quality of the reconstructed signal, a questionnaire as depicted in Fig. 10 is used.

For the semi-blind test, MOS for the k th diagnostic feature of the ECG signal is given as:

$$MOS_k^{SB} = \frac{\sum_{i=1}^N Q(i)}{N} \quad (18)$$

where N is the total number of evaluators and $Q(i)$ is the quality score (out of 5) given by the evaluator for the k th diagnostic feature.

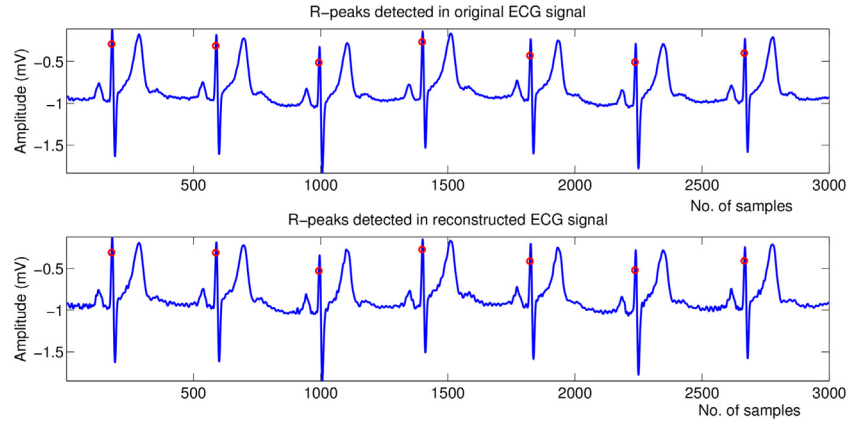


Fig. 6. R-peaks detected in original and reconstructed signals for ECG record no. 117 of the MIT-BIH arrhythmia database.

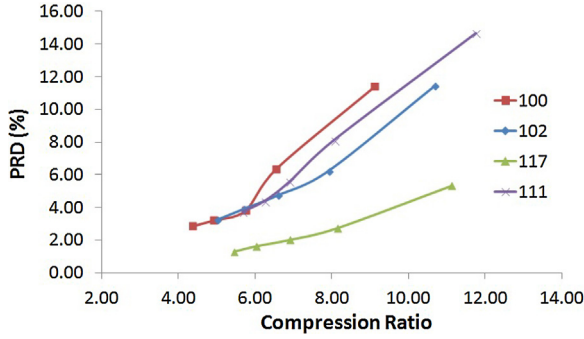


Fig. 7. CR vs PRD curves for different records of ECG signals.

MOS for an ECG signal under test can be defined as the average of the sum of all MOS_k^{SB}

$$MOS_{ECG}^{SB} = \frac{\sum_{k=1}^{N_f} MOS_k^{SB}}{N_f} \quad (19)$$

where N_f is the number of diagnostic features.

For the k th diagnostic feature and overall ECG signal, the gold standard MOS errors (in percentage) are defined as:

$$MOS_k^e = \frac{5 - MOS_k^{SB}}{5} \times 100\% \quad (20)$$

Table 2

MOS errors (in %).

Record no.	MOS errors for different segmentation based features					Overall MOS errors
	P-wave	QRS-complex	T-wave	PR-segment	T-segment	
100	5.00	5.00	8.75	7.50	10.00	7.25
117	3.75	2.5	5.00	3.75	5.00	4.00

$$MOS_{ECG}^e = \frac{5 - MOS_{ECG}^{SB}}{5} \times 100\% \quad (21)$$

where MOS_k^e is MOS error for segmentation based diagnostic features and MOS_{ECG}^e is MOS error for overall ECG signal.

For ECG record no. 100 and 117, MOS errors (in %) for segmentation based diagnostic features as well overall reconstructed ECG signals are demonstrated in Table 2. As per MOS error criteria [34], four ranges of MOS errors are defined: below 15%, 15–35%, 35–50% and above 50% which denotes very good quality, good quality, average quality and unacceptable quality respectively. From Table 2, maximum MOS error is 10% for T-segment of record 100 while it is minimum 2.50% for QRS complex of record 117. The overall MOS errors are 7.25% and 4% for ECG records 100 and 117 respectively. It denotes that clinical qualities of reconstructed ECG signals are fall in the very good range.

Fig. 7 illustrates CR vs PRD curves for different ECG records (including 100 and 117) of the MIT-BIH arrhythmia database. All depicted curves are of similar nature (as CR increases PRD

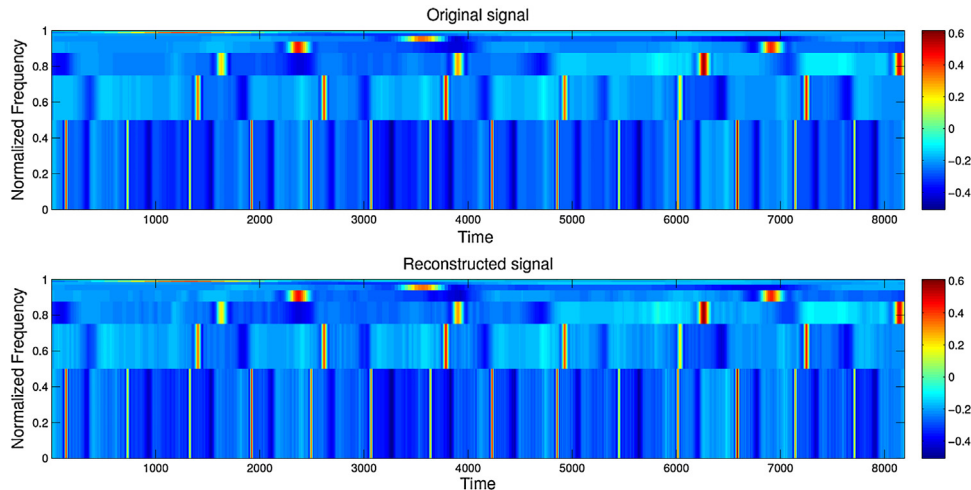


Fig. 8. Normalized frequency vs time plots of original and reconstructed signals (ECG record no. 100).

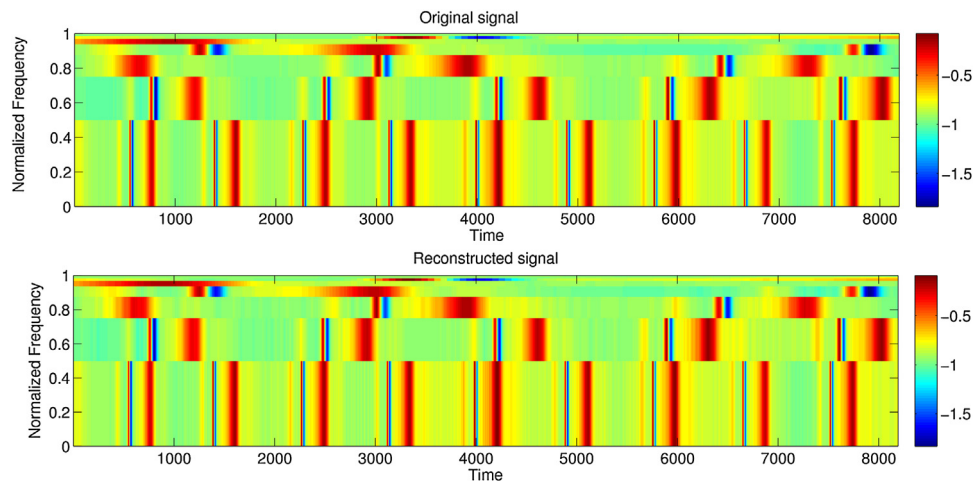


Fig. 9. Normalized frequency vs time plots of original and reconstructed signals (ECG record no. 117).

Database:..... ECG Record No.....	
Evaluator's Name:.....	
Details:.....	
Similarity of segmentation based diagnostic features (score out of 5)	
• P-wave:.....	• QRS-complex:.....
• T-wave:.....	• PR-segment:.....
• ST-segment:.....	
• Overall similarity score:.....	
Comments:.....	

Fig. 10. Semi-blind mean opinion score test.

Table 3
Comparison of performance with other existing methods.

ECG record no.	Algorithm	CR	PRD	QS
100	Lee et al. [17]	4.00	2.30	1.74
	Kumar et al. [12]	5.73	3.21	1.72
	Proposed	5.88	3.81	1.54
117	Hilton et al. [35]	8.00	2.60	3.08
	Djohn et al. [36]	8.00	3.90	2.05
	Kumar et al. [12]	5.65	3.33	1.70
	Proposed	8.13	2.73	2.98

also increases) which assures the working ability of the proposed method on different ECG records.

For ECG record no. 100 and 117, Table 3 provides the comparison of compression performance of the proposed method with existing techniques based on DCT and DWT. In the presented work, DCT is used in the kernel of the DOST. It provides energy compaction to fewer transform coefficients. Therefore, the DCT based DOST associated with DZQ, integer conversion of coefficients and RLE is used to achieve compression on ECG signals. For good comparison with others, performance is evaluated for threshold values of DZQ with $EPE = 99.998\%$. For ECG record no. 100, the proposed method offers higher CR than [17] but higher PRD is also noticed. However, QS is comparable. In comparison with [12], it is found that performance is almost similar. For ECG record no. 117, the proposed method provides better performance (higher CR, lower PRD, higher QS) than [36,12]. In comparison with [35], the proposed method offers almost similar performance in terms of CR,

PRD, and QS. Overall, the compression method presented in this paper provides a competitive performance than existing methods.

6. Conclusion

A novel ECG compression algorithm using DCT based DOST is reported in this paper. In the kernel of the DOST, DCT is used which shows energy compaction property. Further, DZQ, integer conversion of coefficients and run-length encoding help to achieve good compression. The proposed compression algorithm is evaluated using 48 ECG records of the MIT-BIH arrhythmia database. Average CR, PRD, PRDN, QS, RMS, and SNR for all 48 ECG records are 6.27, 5.37, 7.95, 1.49, 0.03 and 22.68 respectively. Compression results of the proposed method exhibit competitive performance than other existing methods. However, compression performance can be improved by applying a non-linear quantization scheme and efficient encoding strategy. To medically validate the proposed method, a subjective measure using a semi-blind MOS test is also performed. From the analysis of MOS test, it is found that diagnostic qualities of reconstructed ECG signals are very good and medically acceptable.

Acknowledgements

We thank Dr. Kamlesh Jha, M.D., Dept. of Physiology, AIIMS Patna, Dr. R.P. Sinha, M.O., IIT Patna Hospital and two researchers (Deba Prasad Dash and Neelam Sharma), Signal and Image Processing Laboratory, Dept. of Electrical Engineering, IIT Patna for their valuable help and suggestions during subjective quality measurement of ECG signals.

References

- [1] S.M. Jaleddine, C.G. Hutchens, R.D. Strattan, W.A. Coberly, ECG data compression techniques – a unified approach, *IEEE Trans. Biomed. Eng.* 37 (4) (1990) 329–343.
- [2] J. Cox, F. Nolle, H. Fozzard, G. Oliver, AZTEC, a preprocessing program for real-time ECG rhythm analysis, *IEEE Trans. Biomed. Eng.* 2 (1968) 128–129.
- [3] V. Kumar, S.C. Saxena, V. Giri, D. Singh, Improved modified AZTEC technique for ECG data compression: Effect of length of parabolic filter on reconstructed signal, *Comput. Electr. Eng.* 31 (4) (2005) 334–344.
- [4] U.E. Ruttimann, H.V. Pipberger, Compression of the ECG by prediction or interpolation and entropy encoding, *IEEE Trans. Biomed. Eng.* 11 (1979) 613–623.
- [5] S. Mukhopadhyay, S. Mitra, M. Mitra, An ECG signal compression technique using ASCII character encoding, *Measurement* 45 (6) (2012) 1651–1660.
- [6] Y. Zigel, A. Cohen, A. Katz, ECG signal compression using analysis by synthesis coding, *IEEE Trans. Biomed. Eng.* 47 (10) (2000) 1308–1316.

- [7] Y. Kocyigit, M. Korurek, B.S. Karlik, B.S. Karlik, ECG Data Compression by Artificial Neural Networks, vol. 1, 1999, pp. 338–339.
- [8] A. Cohen, M. Poluta, R. Scott-Millar, Compression of ECG signals using vector quantization, IEEE South African Symposium on Communications and Signal Processing (COMSIG) (1990) 49–54.
- [9] B.S. Reddy, I. Murthy, ECG data compression using Fourier descriptors, IEEE Trans. Biomed. Eng. 4 (1986) 428–434.
- [10] S. Lee, J. Kim, M. Lee, A real-time ECG data compression and transmission algorithm for an e-health device, IEEE Trans. Biomed. Eng. 58 (9) (2011) 2448–2455.
- [11] W. Kuklinski, Fast Walsh transform data compression algorithm: ECG applications, Med. Biol. Eng. Comput. 21 (4) (1983) 465–472.
- [12] R. Kumar, A. Kumar, R.K. Pandey, Beta wavelet based ECG signal compression using lossless encoding with modified thresholding, Comput. Electr. Eng. 39 (1) (2013) 130–140.
- [13] C.K. Jha, M.H. Kolekar, Efficient ECG data compression and transmission algorithm for telemedicine, International Conference on Communication Systems and Networks (COMSNETS) (2016) 1–6.
- [14] V.A. Motinath, C.K. Jha, M.H. Kolekar, A novel ECG data compression algorithm using best mother wavelet selection, International Conference on Advances in Computing, Communications and Informatics (ICACCI) (2016) 682–686.
- [15] M. Sharafat Hossain, N. Amin, ECG compression using subband thresholding of the wavelet coefficients, Aust. J. Basic Appl. Sci. 5 (5) (2011) 739–749.
- [16] C.K. Jha, M.H. Kolekar, Classification and compression of ECG signal for Holter device, in: Biomedical Signal and Image Processing in Patient Care, IGI Global Publisher, 2018, pp. 46–63.
- [17] H. Lee, K.M. Buckley, ECG data compression using cut and align beats approach and 2-D transforms, IEEE Trans. Biomed. Eng. 46 (5) (1999) 556–564.
- [18] Z. Lu, D.Y. Kim, W.A. Pearlman, Wavelet compression of ECG signals by the set partitioning in hierarchical trees algorithm, IEEE Trans. Biomed. Eng. 47 (7) (2000) 849–856.
- [19] S.-C. Tai, C. Sun, W.-C. Yan, A 2-D ECG compression method based on wavelet transform and modified SPIHT, IEEE Trans. Biomed. Eng. 52 (6) (2005) 999–1008.
- [20] J.-J. Wei, C.-J. Chang, N.-K. Chou, G.-J. Jan, ECG data compression using truncated singular value decomposition, IEEE Trans. Inf. Technol. Biomed. 5 (4) (2001) 290–299.
- [21] C. Zhao, Z. Chen, J. Meng, X. Xiang, Electrocardiograph compression based on sifting process of empirical mode decomposition, Electron. Lett. 52 (9) (2016) 688–690.
- [22] K. Khaldi, A.-O. Boudraa, On signals compression by EMD, Electron. Lett. 48 (21) (2012) 1329–1331.
- [23] G. Grossi, R. Lanzaottili, J. Lin, High-rate compression of ECG signals by an accuracy-driven sparsity model relying on natural basis, Digit. Signal Process. 45 (2015) 96–106.
- [24] R.G. Stockwell, A basis for efficient representation of the S-transform, Digit. Signal Process. 17 (1) (2007) 371–393.
- [25] PhysioNet: the research resource for complex physiological signals, <https://physionet.org/> (accessed 24.03.16).
- [26] J. Ladan, E.R. Vrscaj, The Discrete Orthonormal Stockwell Transform and Variations, with Applications to Image Compression, 2013, pp. 235–244.
- [27] J. Chen, J. Ma, Y. Zhang, X. Shi, ECG compression based on wavelet transform and golomb coding, Electron. Lett. 42 (6) (2006) 322–324.
- [28] B.A. Rajoub, An efficient coding algorithm for the compression of ECG signals using the wavelet transform, IEEE Trans. Biomed. Eng. 49 (4) (2002) 355–362.
- [29] J. Chen, F. Wang, Y. Zhang, X. Shi, ECG compression using uniform scalar dead-zone quantization and conditional entropy coding, Med. Eng. Phys. 30 (4) (2008) 523–530.
- [30] C.K. Jha, M.H. Kolekar, ECG data compression algorithm for tele-monitoring of cardiac patients, Int. J. Telemed. Clin. Pract. 2 (1) (2017) 31–41.
- [31] X. Wang, J. Meng, A 2-D ECG compression algorithm based on wavelet transform and vector quantization, Digit. Signal Process. 18 (2) (2008) 179–188.
- [32] J. Pan, W.J. Tompkins, A real-time QRS detection algorithm, IEEE Trans. Biomed. Eng. 3 (1985) 230–236.
- [33] A.S. Al-Fahoum, Quality assessment of ECG compression techniques using a wavelet-based diagnostic measure, IEEE Trans. Inf. Technol. Biomed. 10 (1) (2006) 182–191.
- [34] Y. Zigel, A. Cohen, A. Katz, The weighted diagnostic distortion (WDD) measure for ECG signal compression, IEEE Trans. Biomed. Eng. 47 (11) (2000) 1422–1430.
- [35] M.L. Hilton, Wavelet and wavelet packet compression of electrocardiograms, IEEE Trans. Biomed. Eng. 44 (5) (1997) 394–402.
- [36] A. Djohan, T.Q. Nguyen, W.J. Tompkins, ECG compression using discrete symmetric wavelet transform, Eng. Med. Biol. Soc. 1 (1995) 167–168.